



Embed2Scale case study

Forest Disturbance Monitoring with Geospatial Foundation Models



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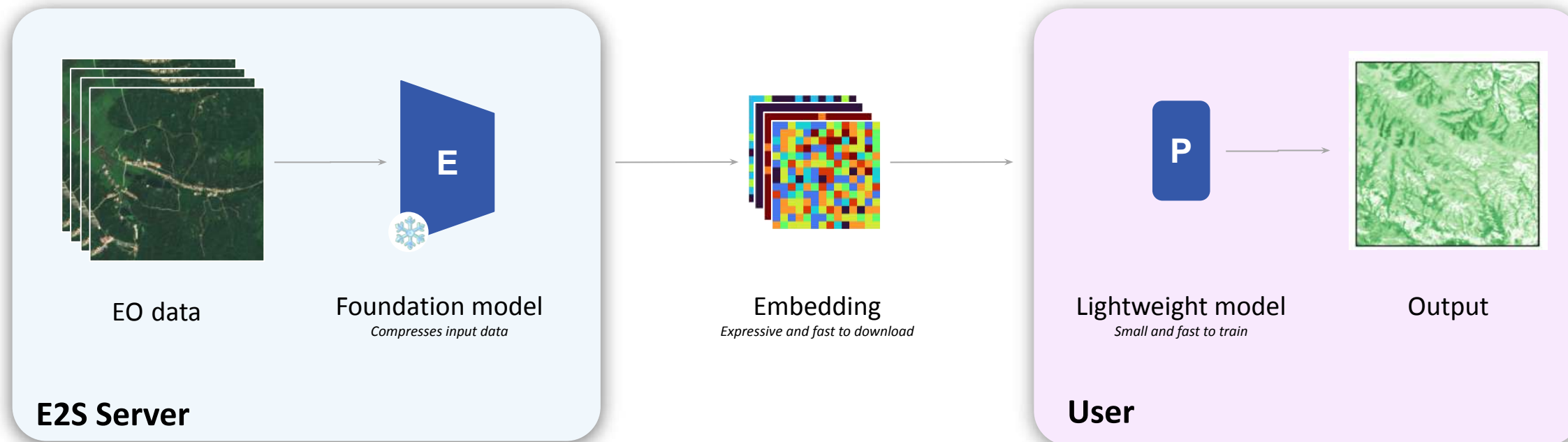


Lowering *expertise* and *hardware barriers* for using *Earth Observation* (EO) data and using “AI” for EO

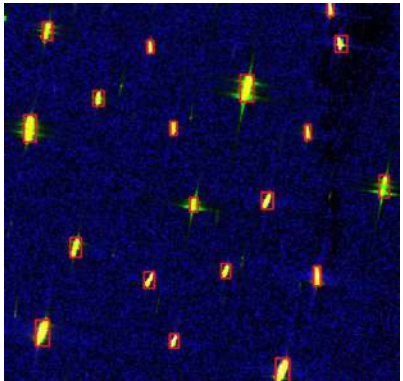


Funding - European Commission, Switzerland & UK

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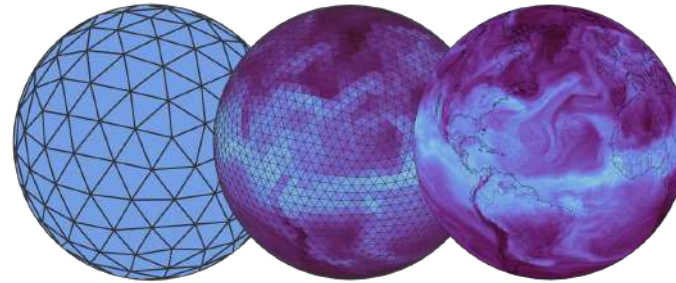
Embed2Scale - Demonstration applications



Maritime awareness



Crop monitoring

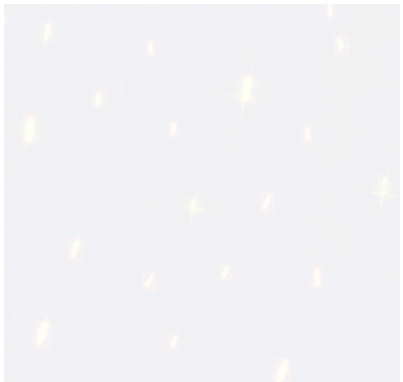


Climate data compression

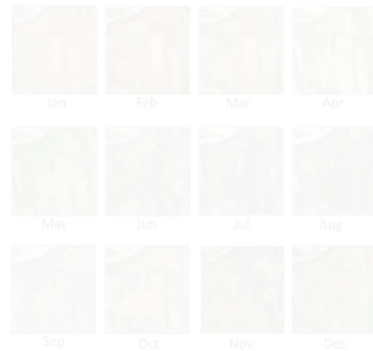


Forest monitoring

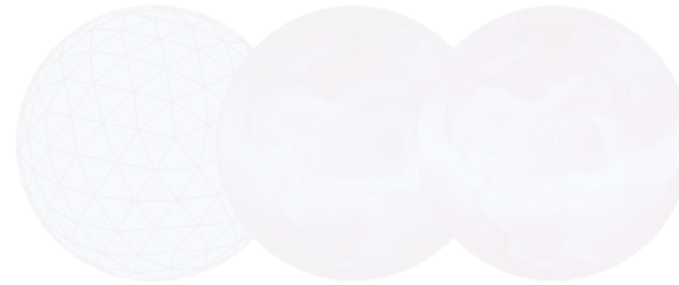
Embed2Scale - Demonstration applications



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Forest monitoring

Forest mapping

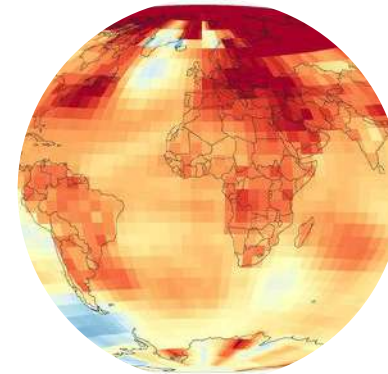




Forest mapping - Motivation



Ecology



Climate science



Timber



Sustainable supply chain

Credit: image 1, image 2, image 3, image 4



Forest mapping - Not so easy

Forest **definition**

Tree cover %, tree height, species, land use

Forests are **dynamic systems**

Natural growth, (un)planned disturbances

Mapping forests is **non-trivial**

Spatiotemporal scale, biomes, sensors, semantics



Credit: Unravelling Europe's forest disturbance dynamics from space, Cornelius Senf, TUM, 2022



Forest disturbance monitoring



Pests



Fire



Windthrow



Logging

In-situ observation accurate but not scalable - need for **EO-based monitoring**

Various event speeds, requiring **daily to monthly monitoring**

Diverse **agents** affecting **tree cover** and **land use** and potentially **correlating**

Require temporal understanding for **change detection** and **agent attribution**

Regional patterns requiring tailored models for local forest practitioners



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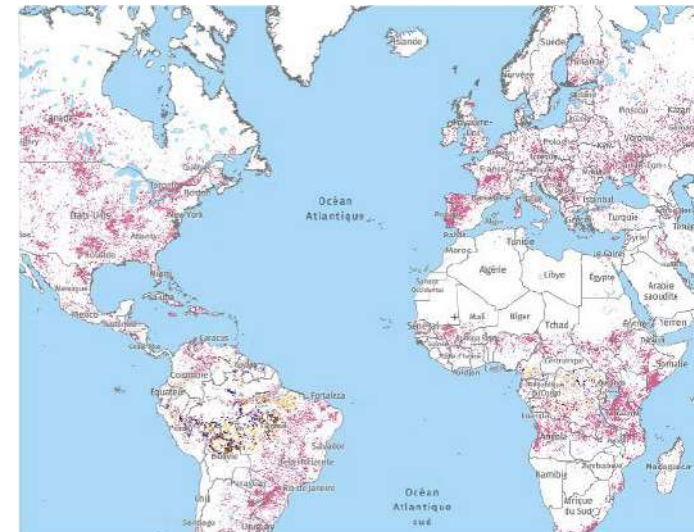
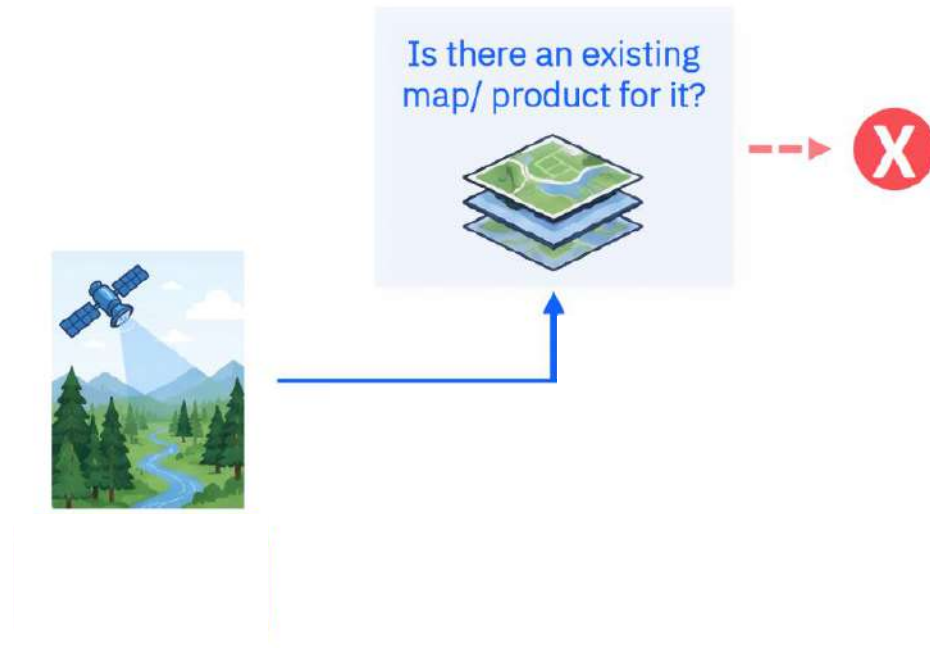




Approach - Existing products

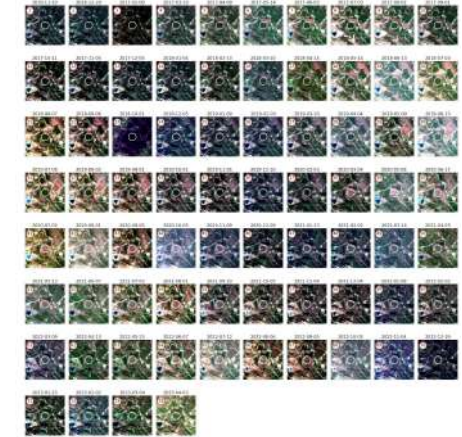
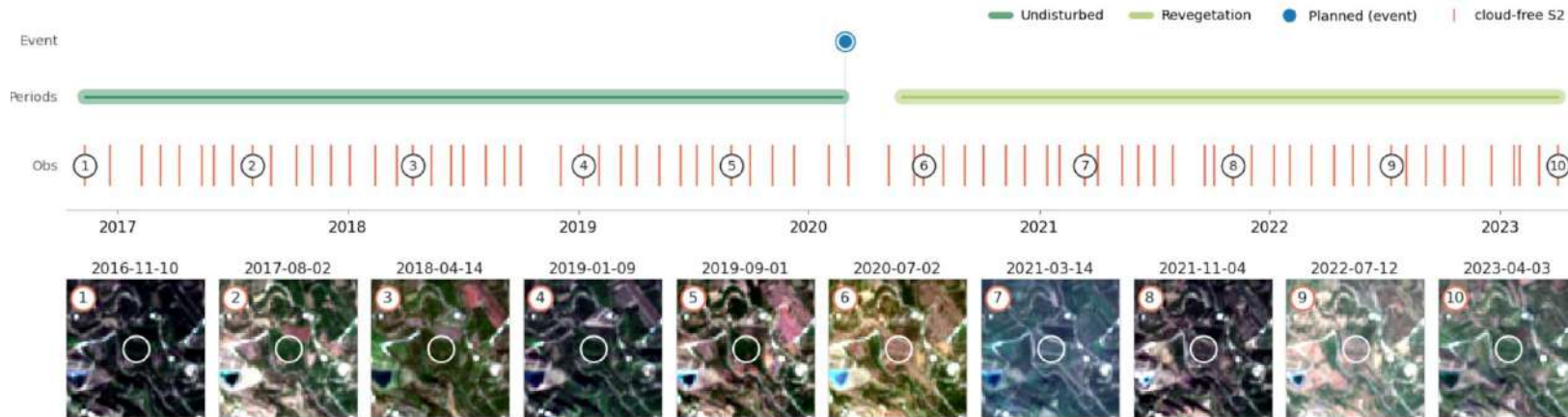
Existing near real-time products

Temporal heuristics | No spatial context | Detection **XOR** attribution



Global Forest Watch
Hansen et al, 2013

Our training dataset



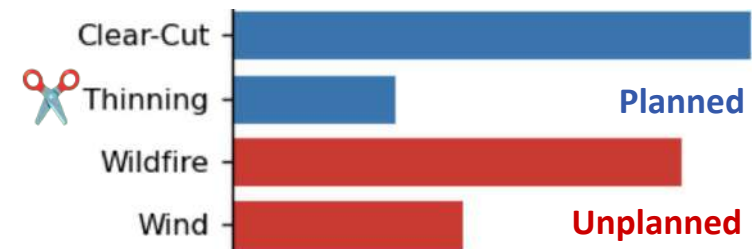
 **200** sample locations

 **2015-2024** time series

 **1 disturbance** event per time series

 **252x252** pixel patches

 **Sentinel-2** L2A cloud-free-ish patches





Approach - Modeling from scratch



Supervised model



Training a custom model from scratch

Limited reference data for learning deep image features

Risk of **overfitting** and **poor generalization**



Approach - EO embedding products



EO embedding products

AlphaEarth - **closed source** generation and **yearly resolution**

TESSERA - **yearly temporal aggregation**





Approach - EO embedding workflow

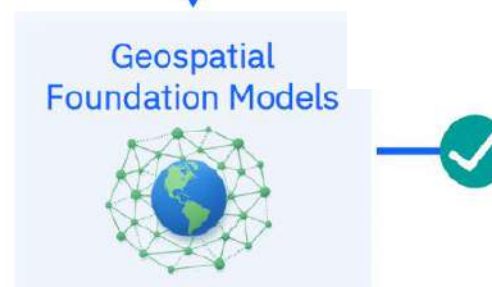
EO embedding workflows

On-demand embeddings for each observation date

Pre-computed embeddings can alleviate redundant operations

Embeddings capture **spatial** and **spectral context**

💡 Search for disturbance signatures in time series of embedding vectors





Embedding-based pipeline

